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Student number : _____

Signature : _____

EXAMINATION OF GENERALIZED LINEAR AND ADDITIVE MODELS

PROFESSOR : EVA CANTONI

AUGUST 30TH, 2022

DURATION : 120 minutes. The duration of the examination is strict. No additional time is allowed for copying down answers.

AUTHORIZED MATERIALS :

- ☐ None - Closed-book exam (only paper bilingual dictionaries are allowed)
- ☒ Personal notes and paper course materials - Open-book exam
- ☒ Non programmable calculator
- ☐ Scientific calculator with empty memory
- ☐ Computer (only for computer-based examinations)
- ☐ Other :

MATERIAL TO BE HANDED IN :

- | | | |
|--|--|--------------------------------------|
| ☒ Exam paper | ☒ Exam sheet | ☐ Answer grid |
| ☐ Draft paper | ☐ Other : | |

Leave the colored card on the table and return any materials you might have borrowed.

INSTRUCTIONS :

- Please write your answers on the official UNIGE colored paper.
- For the plots and outputs analysis, you are allowed to "doodle" the exam copy (exam paper) and reference these remarks on your written answer.
- Please reference figures and outputs properly. For this purpose, we have numbered the figures and numbered the lines in the output listings.
- Statistical tables are provided at the end of the exam.
- Si vous voulez, vous pouvez aussi répondre en français.

In accordance with the examination guidelines, in the event of inappropriate behavior and attempted cheating, the supervisor will note the time and events on the examination paper and inform the teacher responsible for the course, while allowing the student to complete the examination.

Exam

August 30th, 2022

Exercise 1

(36 points)

The dataset `aids` gives the delay between the time a patient is diagnosed with AIDS and the time that this diagnosis is reported to the Communicable Disease Surveillance Centre, as required in England and Wales to monitor the epidemic.

The dataset contains 570 observations with 5 columns:

- `year`: the year of the diagnosis, between July 1983 and December 1992.
- `quarter`: the quarter of the year in which diagnosis was made, from 1 to 4.
- `time`: the time of diagnosis, counted as the number of quarters from July 1983.
- `delay`: the delay, in quarters, between diagnosis and reporting to the Communicable Disease Surveillance Centre. For each `year`, the variable `delay` takes 15 fix values on a grid between 0 and 41 as follow: 0, 2, 5, 8, ..., 41 (steps of 3, except for the the first one).
- `y`: the number of cases reported.

For instance, the 100-th line:

year	quarter	time	delay	y
1985	1	7	26	2

indicates that there were 2 cases of AIDS diagnosed 7 quarters after July 1983 (corresponding to the first quarter of year 1985) which were reported to the Communicable Disease Surveillance Centre with a delay of 26 quarters (i.e. 6 years and a half after the diagnosis).

We would like to model the variable `y` as a function of the others.

We started by fitting a Poisson GLM model, Model 0, as appearing in Output 0.

Output 0: Model 0 (GLM).

```
1 > fit0 <- glm(y ~ year + quarter + time + delay, data = aids, family = poisson())
2 > fit0
3 Call: glm(formula = y ~ year + quarter + time + delay, family = poisson(),
4 data = aids)
5
6 Coefficients:
7 (Intercept)      year      quarter      time      delay
8 -4.232e+02    2.148e-01   -8.497e-03         NA   -1.432e-01
9
10 Degrees of Freedom: 569 Total (i.e. Null); 566 Residual
11 Null Deviance: 16720
12 Residual Deviance: 4253 AIC: 5659
```

1. (2 points) Some NA appear in the estimates of the model shown in Output 0. Why ?

We next fitted another Poisson GLM model, Model 1, see Output 1.

Output 1: Model 1 (GLM).

```
1 > fit1 <- glm(y ~ time + delay, data = aids, family = poisson())
2 > fit1
3
4 Call: glm(formula = y ~ time + delay, family = poisson(), data = aids)
5
6 Coefficients:
7 (Intercept)      time      delay
8 2.76554      0.05318     -0.14316
9
10 Degrees of Freedom: 569 Total (i.e. Null); 567 Residual
11 Null Deviance: 16720
12 Residual Deviance: 4284 AIC: 5687
```

2. (2 points) Interpret the sign of the coefficient associated with the variable `time` in Model 1.
3. (2 points) According to Model 1, estimate the expected number of cases diagnosed at `time = 11` and reported with a delay of 5 quarters.
4. (2 points) Compute the log-likelihood of Model 1.

We also fitted a simpler model, Model 2, using only the variable `time`, see Output 2.

Output 2: Model 2 (GLM).

```
1 > fit2 <- glm(y ~ time, data = aids, family = poisson())
2 > Call: glm(formula = y ~ time, family = poisson(), data = aids)
3
4 Coefficients:
5 (Intercept)      time
6 1.20354      0.05318
7
8 Degrees of Freedom: 569 Total (i.e. Null); 568 Residual
9 Null Deviance: 16720
10 Residual Deviance: 14770 AIC: 16180
```

5. (2 points) Which of Model 1 or Model 2 performs best ? Why ?
6. (2 points) Give the value of the deviance D for Model 1. Perform a goodness-of-fit test at 5% level to test whether the model is good.
7. (2 points) Compute the percentage of deviance explained by Model 1.

Figure 1 shows the diagnostic plots of Model 1. Figure 2 shows $\hat{y}$ as a function of `delay` and Figure 3 shows the distribution of $\hat{y}$.

8. (4 points) Comment on Figure 1, 2 and 3. What potential problems in Model 1 could these figures suggest ? How would you address them ?

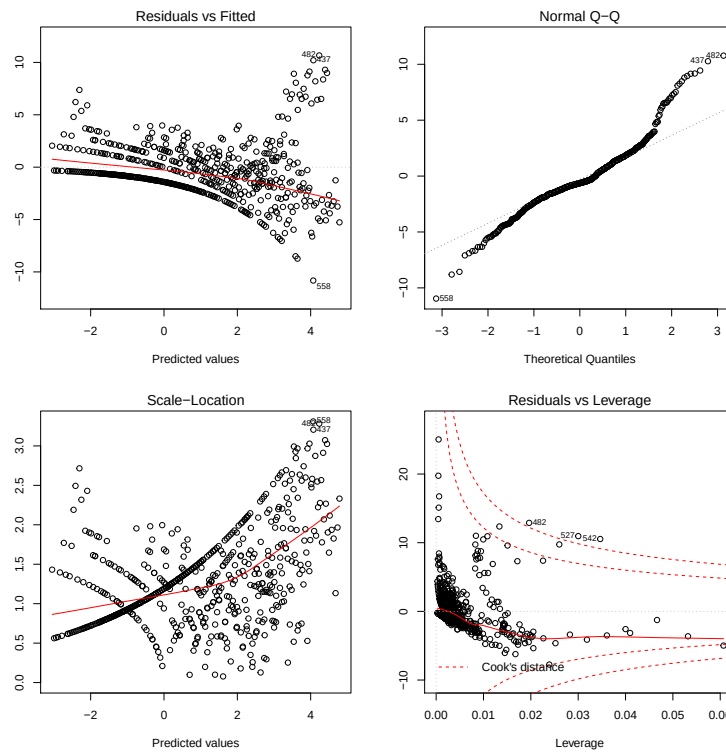


Figure 1: Diagnostic plots of Model 1.

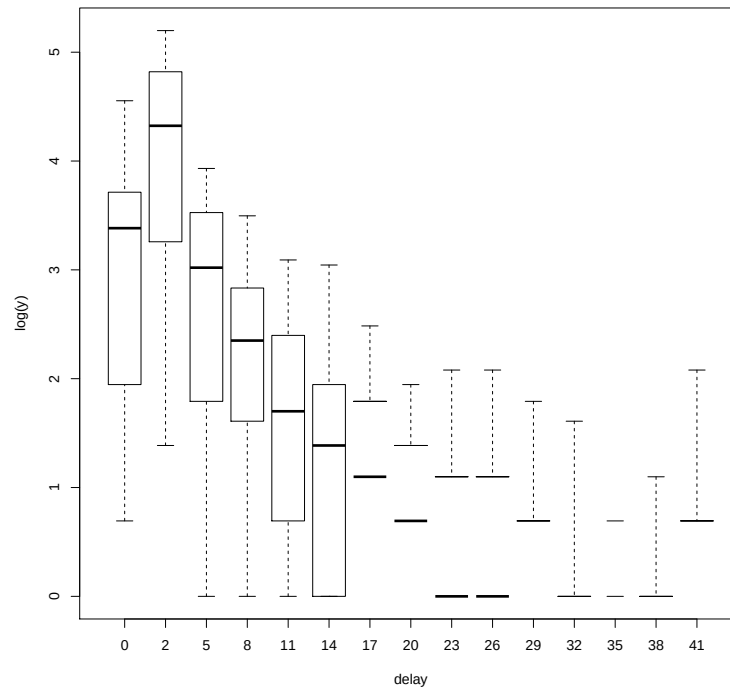


Figure 2: $\log(y)$ as a function of delay.

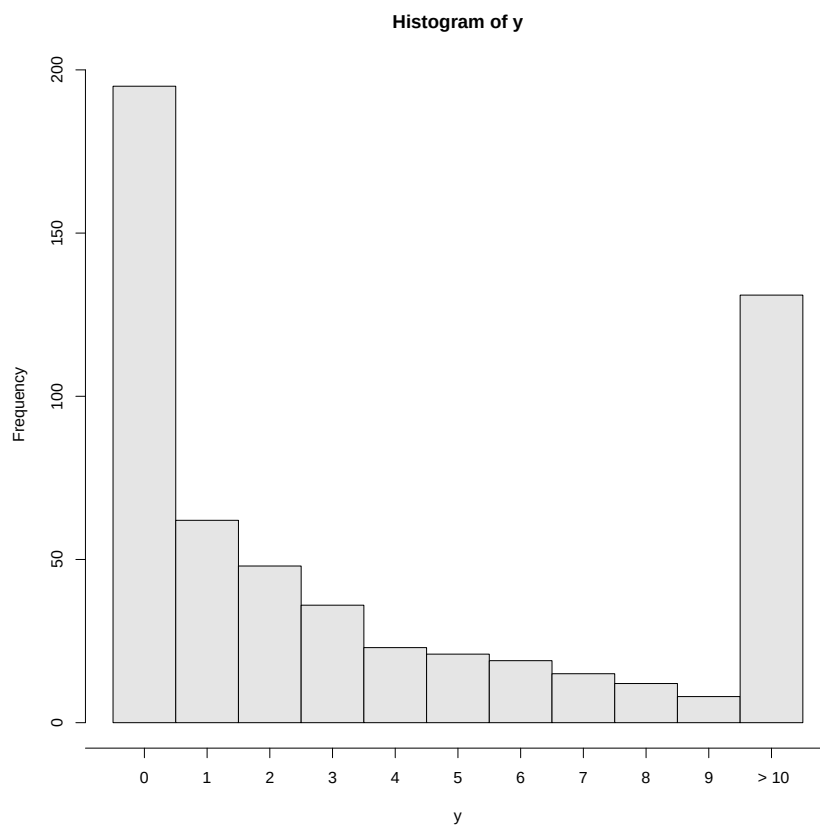


Figure 3: Distribution of y .

We continued the analysis by fitting a GAM model, Model 3, presented in Output 3 and Figure 4.

Output 3: Model 3 (GAM).

```

1 > fit_gam <- gam(y ~ s(time) + s(delay), data = aids, family = poisson)
2 > summary(fit_gam)
3
4 Family: poisson
5 Link function: log
6
7 Formula:
8 y ~ s(time) + s(delay)
9
10 Parametric coefficients:
11 Estimate Std. Error z value Pr(>|z|)
12 (Intercept)  0.88146    0.03859   22.84   <2e-16 ***
13 ---
14 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
15
16 Approximate significance of smooth terms:
17 edf Ref.df Chi.sq p-value
18 s(time)  7.716  8.509   1541   <2e-16 ***
19 s(delay)  8.954  8.999   7440   <2e-16 ***
20 ---
21 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
22
23 R-sq.(adj) =  0.867   Deviance explained = 87.2%
24 UBRE = 2.8257   Scale est. = 1           n = 570
25 AIC = 3578.037
26
27 > gam.check(fit_gam)
28
29 Method: UBRE   Optimizer: outer newton
30 full convergence after 9 iterations.
31 Gradient range [1.312875e-11,4.753834e-06]
32 (score 2.825708 & scale 1).
33 Hessian positive definite, eigenvalue range [0.0001600396,0.00211296].
34 Model rank =  19 / 19
35
36 Basis dimension (k) checking results. Low p-value (k-index<1) may
37 indicate that k is too low, especially if edf is close to k'.
38
39 k'   edf k-index p-value
40 s(time)  9.00 7.72    0.84   <2e-16 ***
41 s(delay)  9.00 8.95    0.42   <2e-16 ***
42 ---
43 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

9. (2 points) Can the AIC of the GAM model be compared to the GLM models' AIC ?
10. (2 points) Is GAM Model 3 more relevant than Model 1 ?
11. (2 points) Is the number of basis functions used for splines high enough ?

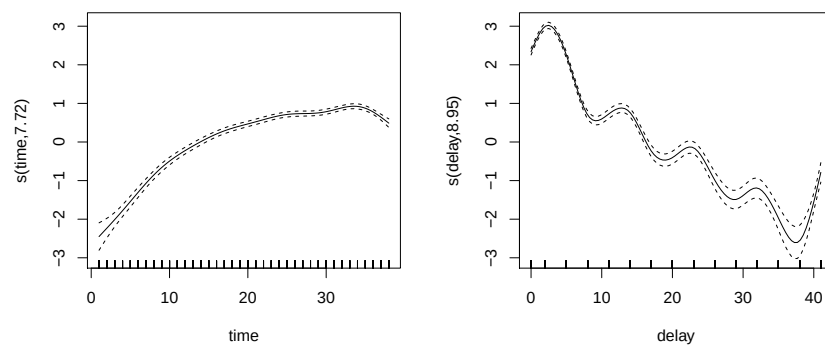


Figure 4: Fitted functions of the GAM Model 3.

We fitted another GAM model, Model 4, presented in Output 4 and Figure 5.

Output 4: Model 4 (GAM).

```

1  > fit_gam_quasi <- gam(y ~ s(time) + s(delay), data = aids, family = quasipoisson)
2  > summary(fit_gam_quasi)
3
4  Family: quasipoisson
5  Link function: log
6
7  Formula:
8  y ~ s(time) + s(delay)
9
10 Parametric coefficients:
11 Estimate Std. Error t value Pr(>|t|)
12 (Intercept)  0.89611    0.06793   13.19  <2e-16 ***
13 ---
14 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
15
16 Approximate significance of smooth terms:
17 edf Ref.df      F p-value
18 s(time)  5.651  6.764  70.28  <2e-16 ***
19 s(delay)  8.827  8.987 255.94  <2e-16 ***
20 ---
21 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
22
23 R-sq.(adj) =  0.865   Deviance explained = 87.1%
24 GCV = 3.9956   Scale est. = 3.2445    n = 570

```

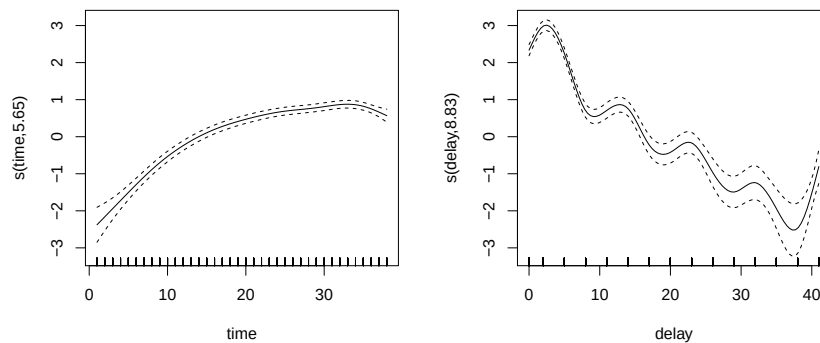


Figure 5: Fitted functions of the GAM Model 4.

12. (2 points) What is the difference between GAM Models 3 and 4 ? Explain.
13. (2 points) Could we compare the AIC of Model 3 with the AIC of Model 4 ?

We now consider the issue of excess of zeros.

14. (2 points) Cite two models that deal with excess of zeros and describe the difference between them.

Output 5 shows a model, Model 5, accounting for excess of zeros.

Output 5: Model 5 (hurdle model).

```
1 > fit3_bin <- glm(I(y > 0) ~ delay + time, data = aids, family = binomial)
2 > fit3_trunc <- glm(y ~ delay + time, data = aids, family = truncpoisson,
3   subset = (y > 0))
4 > fit3_bin
5
6 Call: glm(formula = I(y > 0) ~ delay + time, family = binomial, data = aids)
7
8 Coefficients:
9 (Intercept)      delay      time
10 3.39749      -0.10103     -0.02392
11
12 Degrees of Freedom: 569 Total (i.e. Null); 567 Residual
13 Null Deviance: 732.4
14 Residual Deviance: 571.6 AIC: 577.6
15 > fit3_trunc
16
17 Call: glm(formula = y ~ delay + time, family = truncpoisson, data = aids,
18 subset = (y > 0))
19
20 Coefficients:
21 (Intercept)      delay      time
22 2.54980      -0.12182     0.06106
23
24 Degrees of Freedom: 374 Total (i.e. Null); 372 Residual
25 Null Deviance: 11580
26 Residual Deviance: 2958 AIC: 4098
```

15. (2 points) According to Model 5, what is the probability that no case diagnosed at `time= 11` be reported to the Communicable Disease Surveillance Centre with a delay of 5 quarters ?
16. (2 points) Would it be relevant to compute the AIC of Model 5 by summing the AICs of `fit_bin` and `fit_trunc` presented in Output 5? Why ? Does Model 5 improve on Model 1 ?
17. (2 points) Can the different models fitted previously answer the question "Is the delay between diagnosis and reporting to the Communicable Disease Surveillance Centre decreasing with time ?". If no, how would you proceed to answer it ? Share your reasoning, even if it is not conclusive.

Exercise 2

(24 points)

The Beta distribution is used to model proportions. Assume that a random variable Y follows a Beta distribution, with density

$$f(y; p, q) = \frac{\Gamma(p+q)}{\Gamma(p)\Gamma(q)} y^{p-1} (1-y)^{q-1},$$

where $0 < y < 1$, $p > 0$ and $q > 0$. It holds that

$$E(Y) = \frac{p}{p+q},$$

and

$$Var(Y) = \frac{pq}{(p+q)^2(p+q+1)}.$$

1. (4 points) Consider the following transformation of parameters: $\mu = p/(p+q)$ and $\phi = p+q$. Rewrite the density f as a function of μ and ϕ and give $E(Y)$ and $Var(Y)$ under this parametrization.
2. (2 points) Considering ϕ as a nuisance parameter, does the Beta distribution belongs to the one-parameter exponential family?
3. (2 points) For each individual i of a sample of size n , in addition to the individual's response y_i , we have also measured a set of covariates x_i . We would like to use this information in a regression setting in a GLM way. Suggest a possible model (in particular an appropriate link function), where the set of regression parameters will be denoted as β .
4. (4 points) Write the log-likelihood associated with the model you have defined in 3. as a function of β and ϕ .

We consider an application of the above model to the household food expenditure data from Griffiths et al. (1993). The dataset of sample size 38 contains three variables:

- `food`: food expenditure of the household
- `income`: household income
- `persons`: number of persons in the household

Figure 6 give a representation of the dataset and the fitted model is presented in Output 6.

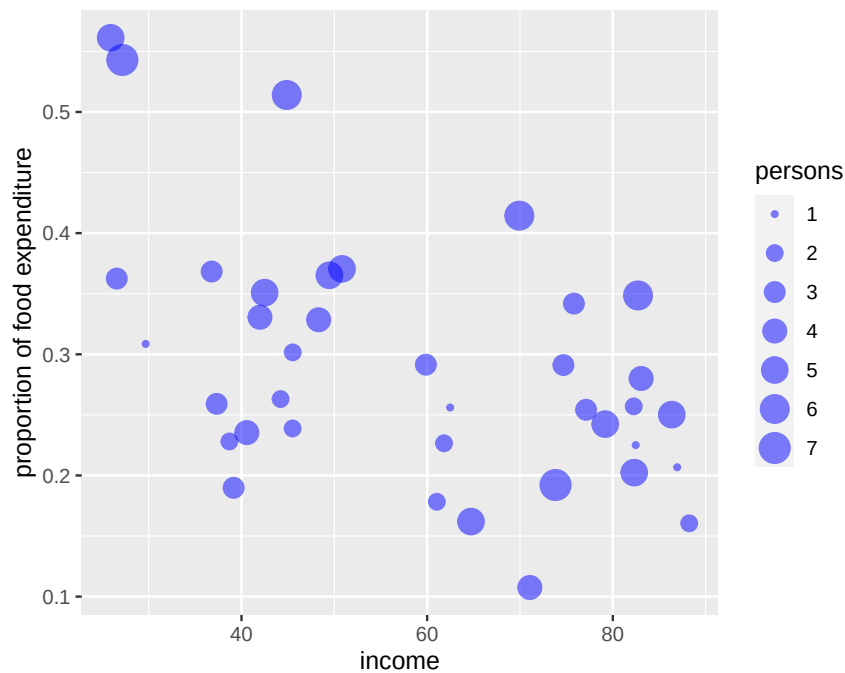


Figure 6: Food expenditure data: proportion of household income spent on food as a function of household income. The size of the dots is proportional to the number of persons in the household.

Output 6: GLM Beta model with logit link.

(Note: the gam function is used to fit the GLM model because it has a Beta family)

```
1 > fit_beta <- gam(I(food/income) ~ income + persons,family=betar, data = FoodExpenditure)
2 > summary(fit_beta)
3
4 Family: Beta regression(33.032)
5 Link function: logit
6
7 Formula:
8 I(food/income) ~ income + persons
9
10 Parametric coefficients:
11             Estimate Std. Error z value Pr(>|z|)
12 (Intercept) -0.621147  0.231921 -2.678  0.00740 **
13 income      -0.012254  0.003142 -3.900  9.64e-05 ***
14 persons      0.118022  0.036595  3.225  0.00126 **
15 ---
16 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
17
18 R-sq.(adj) =  0.438   Deviance explained = 41.6%
19 -REML = -36.2   Scale est. = 1          n = 38
20 > logLik(fit_beta)
21 'log Lik.' 45.28002 (df=....)
```

5. (2 points) What is the effect of increasing the number of persons in the household on the expected proportion of food expenditure?

6. (4 points) A household with 4 persons and an income equal to 60 is planning to welcome an Ukrainian refugee. Quantify the expected proportion of food expenditure change.
7. (2 points) Compute the AIC of the Beta model in Output 6.

We also fitted a GAM model. Output 7 gives the details of the fitted model and Figure 7 depicts the nonparametric fitted functions.

Output 7: GAM Beta model with logit link.

```

1 > fit_beta_gam <- gam(I(food/income) ~ s(income) + s(persons,k=7), family=betar,
2   data = FoodExpenditure)
3
4 > summary(fit_beta_gam)
5
6 Family: Beta regression(34.779)
7 Link function: logit
8
9 Formula:
10 I(food/income) ~ s(income) + s(persons, k = 7)
11
12 Parametric coefficients:
13             Estimate Std. Error z value Pr(>|z|)
14 (Intercept) -0.91450    0.06034  -15.15   <2e-16 ***
15 ---
16 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
17
18 Approximate significance of smooth terms:
19             edf Ref.df Chi.sq  p-value
20 s(income)   1.851   2.312  17.94 0.000257 ***
21 s(persons)  1.000   1.000  10.54 0.001168 **
22 ---
23 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
24
25 R-sq.(adj) =  0.467   Deviance explained =  46%
26 -REML = -40.009   Scale est. = 1           n = 38
27
28 > AIC(fit_beta_gam)
29 [1] -82.79725

```

8. (2 points) What is the argument k in $s(\text{persons}, k = 7)$ in Output 7? Why has it been set to 7?
9. (2 points) Are the nonparametric fits worth the trouble?

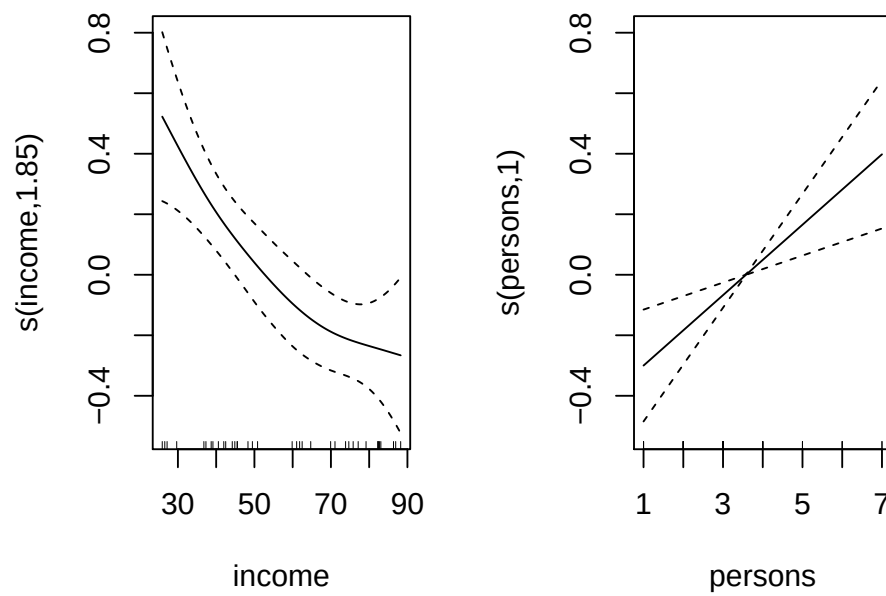
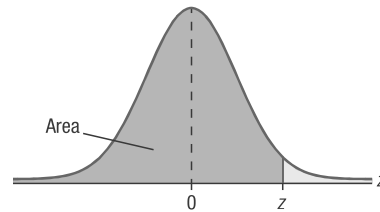


Figure 7: Fitted functions of the GAM model.

B Standard Normal Distribution

Numerical entries represent the probability that a standard normal

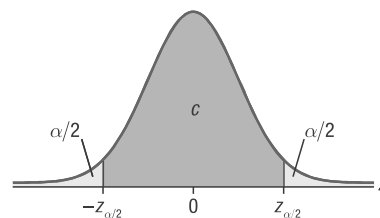
random variable is between $-\infty$ and z where $z = \frac{x - \mu}{\sigma}$.



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Critical Values of z

Level of Confidence, c	$\alpha = 1 - c$	$z_{\alpha/2}$
0.80	0.20	1.28
0.85	0.15	1.44
0.90	0.10	1.645
0.95	0.05	1.96
0.98	0.02	2.33
0.99	0.01	2.575



df	0.75	0.90	0.95	0.975	0.99
1	1.323	2.706	3.841	5.024	6.635
40	45.62	51.81	55.76	59.34	63.69
80	88.13	96.58	101.9	106.6	112.3
120	130.1	140.2	146.6	152.2	159.0
160	171.7	183.3	190.5	196.9	204.5
200	213.1	226.0	234.0	241.1	249.4
240	254.4	268.5	277.1	284.8	293.9
280	295.6	310.7	320.0	328.2	338.0
320	336.7	352.8	362.7	371.4	381.8
360	377.7	394.8	405.2	414.5	425.3
400	418.7	436.6	447.6	457.3	468.7
440	459.6	478.4	489.9	500.0	511.9
480	500.5	520.1	532.1	542.6	555.0
520	541.4	561.7	574.2	585.1	597.9
560	582.2	603.3	616.2	627.5	640.8
600	623.0	644.8	658.1	669.8	683.5
640	663.8	686.3	700.0	712.0	726.2
680	704.5	727.7	741.8	754.2	768.7
720	745.2	769.0	783.5	796.3	811.2
760	785.9	810.4	825.2	838.3	853.6

Table 1: Quantiles of a χ^2 distribution, depending on the degrees of freedom df, for probabilities 0.75, 0.9, 0.95, 0.975 and 0.99